parts of the same polypeptide or two different polypeptides together and are the only covalent interactions involved in the formation of tertiary structure.

3.3.4 Quaternary structure of proteins

Proteins that are made up of a single polypeptide chain have only three levels of structure. Some proteins are made up of more than one polypeptide chain. In such cases the tertiary structures formed by each of those polypeptide chains come together to form a quaternary structure. These individual polypeptide chains are also known as subunits. Hemoglobin, a protein which carries oxygen in blood is made up of four subunits. Similarly, DNA polymerase, an enzyme which synthesizes new strands of DNA is composed of ten subunits. The same types of interactions that contribute to tertiary structure are also involved in stabilization of the quaternary structure.

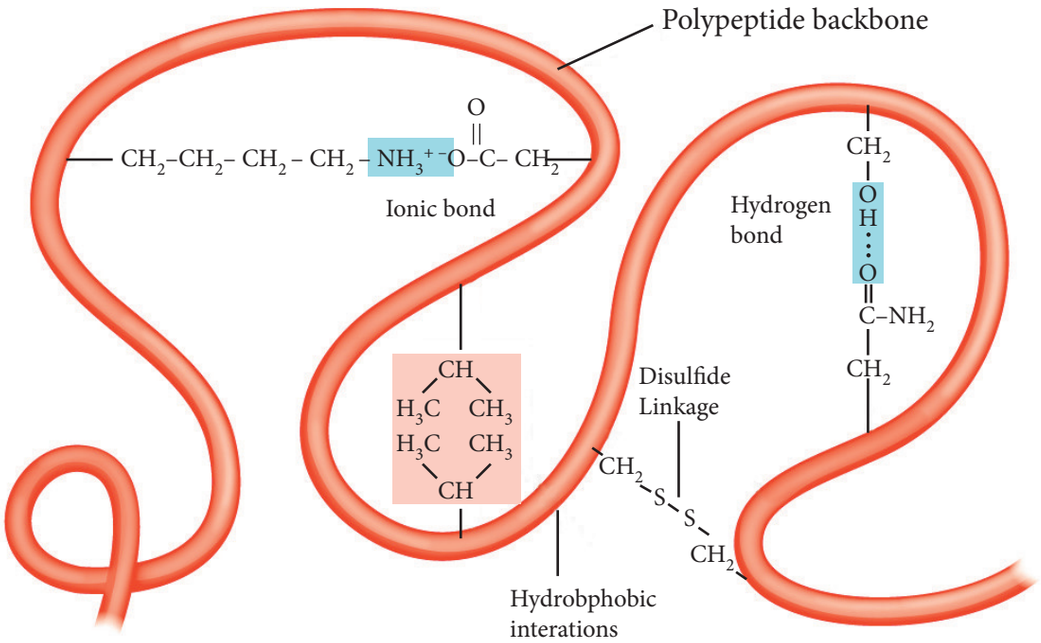


Figure 3.12: molecular interactions in a tertiary protein structure

3.4 Properties of proteins

a. Generally proteins are colourless and tasteless. However, there are exceptions for example Hemoglobin is red in colour.

b. The solubility of proteins is influenced by pH. Proteins are least soluble at their iso electric point.

c. All protein solutions are optically active. The magnitude of optical activity depends on temperature, the wave length of light used and the concentration of protein.

Unit 3.indd 90 21-05-2018 15:09:33